

BTS-V(S)-04.18-0780

Reg.No.

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B

B.Tech. Degree V Semester Supplementary Examination
April 2018

ME 15 - 1506 POWER PLANT ENGINEERING
(2015 Scheme)

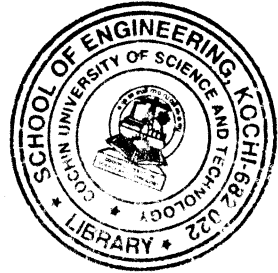
Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define Demand Factor and Load Factor of a Power Plant.
(b) List the factors which should be considered while designing a power plant.
(c) What is a surge tank and name any two types of surge tanks?
(d) What factors should be considered while selecting a site for diesel power plant?
(e) Enumerate the merits of closed cycle gas turbine cycle over open cycle gas turbine.
(f) Briefly explain Fluidized Bed Combustion (FBC).
(g) How are dust collectors classified?
(h) Enumerate any six desirable characteristics of a nuclear coolant.
(i) What are the advantages and disadvantages of geothermal energy?
(j) Explain the principle of fuel cells.



PART B

(4 × 10 = 40)

- II. (a) Briefly explain various costs elements to form total cost of a power system. (5)
(b) A generating station supplies the following loads : 15 MW, 12 MW, 8.5 MW, 5 MW and 0.45 MW. The station has a maximum demand of 22 MW. The annual load factor is 0.48. Calculate (i) The diversity factor and (ii) The demand factor. (5)

OR

- III. (a) Give a brief account of essential components of a hydro electric power plant. (5)
(b) What is meant by a draft tube? Give a brief explanation on any two types of draft tubes. (5)

(P.T.O.)

- IV. With the help of a block diagram explain the essential components and the working of a Diesel engine power plant. (10)
- OR**
- V. (a) What are the advantages of gas turbine power plant over Diesel engine power plant? (5)
- (b) What are the methods for improvement of thermal efficiency of an open cycle gas turbine power plant? Explain how intercooling affects thermal efficiency. (5)
- VI. (a) Explain the principle of operation of overfeed stokers and underfeed stokers. (5)
- (b) Give a detailed explanation of liquid metal cooled nuclear reactors. (5)
- OR**
- VII. (a) What are the different types of steam condensers? Give a brief description on evaporative type of condensers. (5)
- (b) What are the essential components of a nuclear reactor? Give a brief description of reactor core. (5)
- VIII. (a) Give a brief explanation of solar pond technology. (5)
- (b) Give a brief account of any two types of wind turbines. (5)
- OR**
- IX. (a) With neat sketch explain the working of a magneto hydro dynamic power generation. (5)
- (b) Explain the working principle of ocean thermal energy conversion. (5)
